

Chapter 15 – WHEN TRADECRAFT WON'T WORK: DESCRIBING THE BOUNDS FOR ANALYTIC TECHNIQUE

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I suppose it is tempting, if the only tool you have is a hammer, to treat everything as if it were a nail.

Abraham Maslow

When a student of intelligence learns or becomes expert with a particular analytic technique, it is tempting to apply that technique to new intelligence questions that comes along. However, while the hammer is good for problems that are nails, they are less suited for screws, and no use at all on wing nuts. After learning ACH, I often tried to apply the technique in real-world problems. I sometimes became frustrated when the technique wasn't useful. In one instance an explosion had occurred in Indonesia. There was uncertainty as to the cause. Was it terrorism, or an accident? I immediately drew up my ACH matrix and began to enter the pieces of evidence as they were reported. No other explosions or gunfire were noted. The target did not have an obvious political value. The explosion seemed too big to be a gas leak. Terrorist groups had been threatening an attack. There was a history of terrorists setting off explosions of this sort. Additional hypotheses developed. Was it criminal activity? A government attack against a known terrorist location? The news networks brought on terrorism experts who each had at least a dozen new observations and opinions. My ACH matrix quickly became untenable. There was too much evidence of too little quality to make ACH a useful tool. ACH may not be much use at all (it lacks empirical support). Another reason for failure may be that we apply the technique in the wrong situation. It's reasonable to assume a particular Structured Analytic Technique (SAT) is most useful for a particular subset of problems. Hence, it may be helpful to develop a taxonomy that simply describes what families of techniques are useful for particular flavours of problems. To continue the analogy of tools: hammers are effective for pounding problems but not for twisting problems.

The goal of this chapter is to present a classification that assists in mapping analytic techniques to particular intelligence problems. Classification consists of grouping entities according to similarity on some set of criteria. As Bailey notes, the process is so central and ubiquitous to our everyday lives, that we are generally unaware of its existence. And yet, classification lays the foundation for conceptualization and language on a given topic. Without it, there can be no advanced conceptualization, reasoning, language, or data analysis. The secret to effective classification is to identify the fundamental characteristics to base a taxonomy or typology [1]. However, as Heuer and Pherson note, "there is no single right way to organize a taxonomy – only different ways that are more or less useful in achieving a specified goal" [2].

To be fair, most SATs come with a description for what set of circumstances they are best suited for. However, instructors and practitioners of intelligence analysis have a growing list of techniques: ACH, alternative futures, SWOT, Devil's Advocacy, utility matrices, Delphi technique, pre-mortem, etc. The ever-increasing list has grown extensive and is likely intimidating to new students, and burdensome to older analysts. As the number of techniques climbs, practitioners begin to specialize in a subset of techniques they have grown to appreciate and become expert at. The temptation grows to over-apply tools we know best. A taxonomy which maps techniques to problems serves to corral the unstructured list of techniques into a more digestible framework. It reduces complexity, highlights when a technique is unlikely to work, and suggests other techniques that are better suited.

There already exist typologies which classify both intelligence problems and analytic techniques. Treverton and Gabbard divide intelligence problems into puzzles and mysteries. Puzzles have a knowable answer that

can be determined once the correct information is obtained. A mystery is contingent on future influences, and cannot be determined at present with certainty [3]. Rittel and Webber made a formal distinction between tame problems, which scientific methods are largely successful at solving, and wicked problems that do not have a definitive solution, cannot be understood with methods of science, and are largely unique and one-shot affairs [4]. Friedrich Hayek introduced the concept of complex systems. According to Hayek, patterns can be discerned in complex environments, but not precise predictions as is possible with non-complex systems [5]. Most in the intelligence community have heard of Nassim Taleb's "Black Swan" problems, where improbable events can render prediction in complex environments an impossible task [6].

Taxonomies for SATs, or more generally problem-solving techniques, are less common, but still exist. Heuer and Pherson organize SATs in to eight categories according to analyst needs and the cognitive purpose they serve:

- 1) Decomposition and visualization;
- 2) Idea generation;
- 3) Scenarios and indicators;
- 4) Hypothesis generation and testing;
- 5) Assessment of cause-and-effect;
- 6) Challenge analysis;
- 7) Conflict management; and
- 8) Decision support [2].

Guilford makes a distinction between divergent and convergent thinking [7]. Divergent tasks (e.g., how many uses are there for a brick?) can have many acceptable answers. Imagination techniques, or challenge techniques, for example, are techniques designed to increase the available number of potential hypotheses. Hypothesis generation is a divergent thinking task, and analytic tradecraft techniques (e.g., alternative futures, brainstorming, red teaming) have been developed and categorized to assist with this kind of creative thinking. Convergent thinking tasks are those that have a single correct answer (e.g., $2 + 3 = 5$). Selecting the single correct (or at least most likely) hypothesis from a list of possibilities is a convergent thinking task, and many analytic techniques have been developed to assist the analyst in choosing the single most likely option from a list of potential assessments (e.g., Team A / Team B, ACH, Bayesian Belief Networks, Prediction Markets).

There is nothing wrong with any of these taxonomies. All are useful in understanding the problem in some way. However, they don't specifically address the issue of matching tools of intelligence analysis to the intelligence questions based on the nature of the intelligence question. By "nature of the intelligence question" I am referring to the characteristics of the domain the intelligence question exists in (e.g., well understood problem with lots of data), rather than the goal of the technique (e.g., generate hypotheses), which is a more common way to classify techniques.

Snowden and Boone come closest to addressing the fit between nature of a problem and appropriate response [8]. In particular, they address leadership response. Managers must match their style to the complexity of the circumstances they face. Their Cynefin framework (named after the Welsh word for 'habitat') divides circumstances into four contexts: chaotic, complex, complicated, and simple. Managerial best practices differ depending on the domain. Chaotic circumstances, characterized by high turbulence and no clear cause-and-effect relationships, require a leader who takes immediate action to re-establish order, implementing what works as opposed to seeking right answers. When initial turbulence recedes, a chaotic environment may become a complex environment. Containing the characteristics of complex adaptive systems, cause-and-effect relationships are constantly changing. It may be impossible to find a "right"

answer, but with patience it may be possible to detect emerging patterns. If these patterns stabilize, a complex environment becomes a complicated one. A complicated context is characterized by stable cause-and-effect relationships not easily apparent, requiring experts to diagnose the situation. Appropriate manager behaviour includes creating panels of experts and listening to conflicting advice. When the cause-and-effect relationships are better understood, the situation may become a simple environment, with repeating patterns and consistent events. Leaders can apply fact-based management. But beware complacency. The old rules work until they don't work. When the old rules no longer apply, leaders may find themselves thrust back into the chaotic context. Thus, there is a cyclical aspect to this, although not all domains are capable of completing the cycle, as their progress is limited by both our understanding of the domain, as well as the characteristics of the domain itself. When trying to understand tribal allegiances, for instance, it may be that allegiances are constantly changing, and because of that the domain can never evolve beyond a complex context.

Just as Snowden and Boone use the context of a domain to suggest appropriate leadership behaviour, I believe the context can also suggest appropriate focus for the analyst. Prior to becoming aware of Snowden and Boone's work, I defined the context of the environment in terms of the value of theory and the value of data to answer questions in a particular domain. These two criteria, theory and data, form the horizontal and vertical axes of a framework denoting high/low theory application and high/low data application (Figure 15-1). I believe this categorization corresponds well to Snowden and Boone's Cynefin framework, where low theory / low data corresponds to the chaotic context, low theory / high data corresponds to the complex, high theory / high data to complicated, and lastly high theory / low data to the simple context.

Theory/Data Framework for Research & Analysis

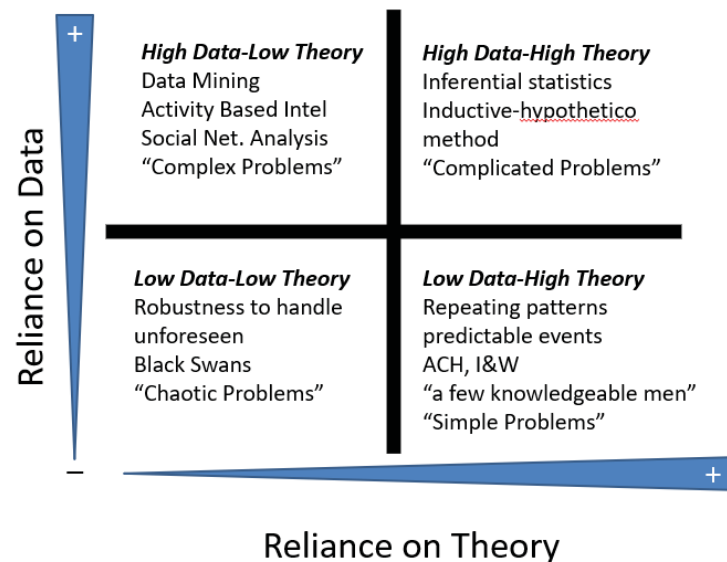


Figure 15-1: Theory/Data Framework for Research and Analysis.

Theory, in this chapter, is being used in the context of how the scientific community uses the word. Theory refers to some sort of organizing principle that explains the relationship between different variables. It is a system of ideas or a set of principles, often dealing with mechanisms or underlying reasons for behaviour, that help us organize and assimilate the observations that we observe. In this usage, a theory is more than "just an idea" [9]. A scientific theory "is grounded in actual data from prior research as well as numerous hypotheses that are consistent with the theory" [9]. Theories are general abstract statements that help to

explain the world around us. A theory is useful in contributing to understanding and knowledge, according to Pettigrew, in at least five important ways. Theories allow us to classify things. These “things” could be entities, processes, or causal relationships. Theories help us predict future events. They help us to understand what causes past events. Finally, they help guide research by providing focus on what questions should be explored more and what can be ignored [10]. A well-established theory is useful in explaining a great number of observable facts [9]. A theory is a general relationship that is relatively generalizable to a wide variety of circumstances. A theory that explains observations in one circumstance, to be useful, can also be used to explain other similar circumstances. Some intelligence questions may be unique one-of-a-kind events. Theory is less helpful in these cases. Other intelligence questions may be able to make better use of past similar events, or utilize general causal models to help predict what might occur. A model inevitably simplifies the real world by ignoring specific details or factors. However, models can have substantial value if they increase our understanding of the observations we make.

Data, as used in this chapter, is simply information and observations. Those observations might be quantitative or qualitative in nature. Some intelligence questions may have data available to draw upon. The explosion of social media and other online resources means that we are practically drowning in data for certain intelligence questions, and the challenge becomes getting and managing the useful data. Other intelligence questions may have relatively little data to draw upon because it is a recent, unfolding event or for some other reason has little information available.

The following examples illustrate how assumptions of theory and data can be used to understand different analytic approaches:

- 1) **Low Reliance on Theory, Low Reliance on Data:** Nasim Taleb’s book, *The Black Swan*, fits well into this quadrant [6]. The Black Swan is a metaphor for a rare event that surprises people, and has a major impact. In hindsight, it is often rationalized as being inevitable or foreseeable. But, in actuality, such low-probability events are not predictable. When we make forecasts, we are making an estimate on what is most likely to occur. Black swan events are, by definition, improbable events. Things like the invention of the personal computer, the internet, the creation of nuclear weapons, the 11 September attacks, World War I... all of these things, Taleb argues, were highly improbable events that were essentially impossible to predict and it’s futile to even try to predict these kinds of things.

To Taleb’s observation I’d like to add something I call the snowflake argument. The idea here is that a particular event can only be understood in the context of the local environment, and it’s useless to try and apply what has happened elsewhere in time or place to the current set of circumstances. Each event is unique, like a snowflake, the likes of which has never been seen before, and never will be seen again. In this notion of knowledge, theory is not highly valued and data is not highly valued. If every event is unique, then nothing you know about the past is going to inform you about the present. And nothing you know about the present is going to inform you about the future. Theory concerns itself with the mechanisms that are generalizable from situation to situation. Theory predicts what is likely to happen. The Black Swan event, by definition, is something that’s not likely to happen. Data might be useful only in retrospect, when determining the impact of a unique event, but it can’t be used for prediction or to develop a more generalizable understanding of concepts.

If you are an intelligence analyst, or an academic researcher, then this quadrant is a depressing quadrant to be in. There is little hope for increasing our understanding. There’s little hope to advance knowledge, because each new event is a completely new set of circumstances. So, what analytic methods are good for answering analytic questions that are in this quadrant?

If we find ourselves in Cynefin’s chaotic context (which I suggest is analogous to the low theory / low data quadrant), Snowden and Boone suggest a limited opportunity for thoughtful analysis. Many quick decisions will need to be made with little time for consideration. Snowden and Boone also note chaos can be an opportunity to implement innovative changes in a system. Analysts

might highlight such opportunities to decision makers. Ultimately the goal is abating the immediate crisis so one can move from a chaotic context to a complex context. This requires satisficing. “Doing what works” quickly is better than “doing what’s right” slowly.

Taleb recommends the only response to a Black Swan event is to develop robust systems that are capable of rebounding to a shock. If an analyst is asked to prepare an analysis on the next Black Swan event, the appropriate response might be to look inward, and evaluate the critical points of failure in one’s own organization. Rather than trying to predict or understand the nature of a future shock event, (e.g., will the price of oil spike next year?), the analyst instead suggests ‘what if’ scenarios that the organization would need to respond to (what if oil suddenly spiked?) and the organization can reflect on how they would be affected. If significant weaknesses are uncovered, systems can be modified to make them more robust to future unpredicted shocks.

- 2) **Low Reliance on Theory, High Reliance on Data:** This quadrant is analogous to Snowden and Boone’s complex context. Theory may be lacking because the domain is too new to be well understood. Or characteristics of the environment mean cause-and-effect relationships are constantly changing, as is often the case in complex adaptive systems, making stable, generalizable theories rare. Some domains may never progress beyond complexity due to the characteristics of the environment. In such cases the search is for stable patterns that will likely have temporary usefulness.

What methods might be useful in this domain? There are data-centric, non-theoretical approaches. One is Activity-Based Intelligence (ABI). Instead of focusing collection on specific objects over a small period of time, this intelligence methodology focuses on collecting data on activities and transactions over long periods of time and large areas [11]. If you build a picture of what “normal” activity looks like, you can spot when something abnormal occurs, and cue a human analyst to take a look at that particular event more closely. Machine learning and data mining can be a data-centric approach as well, where one goes into the maelstrom of data looking for relationships without a theory to guide one’s search.

There is opportunity for a data-centric approach to go wrong. A caution to the data-centric approach is presented by the story of Paul the Octopus. Paul the Octopus lived in an aquarium in Germany and during soccer matches in 2008 and 2010 Paul made predictions of which team would win by eating tasty octopus treats out of one of two possible boxes. He was correct 11 out of 13 times for an impressive success rate of 85%! The oracle Octopus even beat Goldman Sachs analysts who used data to make their own predictions [12]. However, if Paul the Octopus made a new prediction for a future game, I personally would still not place a bet based on his input. Paul may have a good record of picking the winner, but there’s no theoretical reason why Paul’s choice should be any better than a coin flip. Paul the Octopus highlights one of the dangers of a purely data-centric approach. If you look at a large number of variables, some of them will correlate with each other just by chance. If you look at enough octopi, some of them will pick winners.

Taleb [13] highlights this danger: If you take a large number of completely random variables, some of those random variables will correlate just by chance. A set of 200 variables with 1,000 observations each will, for example, almost certainly have at least a few significant correlations. These chance correlations, or spurious correlations, are one of the dangers of using a data-centric approach that does not take into account theory. As you increase the number of variables (even random variables) the more chance correlations you will find.

If patterns can be found and these patterns are relatively stable, they can be used to create theories. These theories can be applied to understand other times, places, and situations. If this is the case, we can progress to a high theory / high data environment.

- 3) **High Reliance on Theory, High Reliance on Data:** If both theory and data can be leveraged, we are in a quadrant analogous to Snowden and Boone's complicated context. Cause-and-effect relationships are relatively stable, but not obvious. Experts are required to uncover the connections. This is close to the traditional scientific method and we have developed excellent methods for this context.

But how might this apply to problems in intelligence? As an illustrative example, let's say you deploy to Afghanistan. While there, some of the local roads get paved. You notice that the insurgency gets worse after the roads are paved. You make that observation and come to a belief that paving roads makes insurgencies worse. You took your Afghanistan experience (data), and you believe you learned something about all insurgencies in general (theory). We can take theory and attempt to validate it with data. Let's take the effect of paved roads on insurgency. We could test that theory by looking at other insurgencies. Our theory makes a prediction that the insurgency will be worse in those areas that have paved roads.

This process is the hypothetico-deductive model, familiar to most scientists. We start out with an observation such as our deployment to Afghanistan. We use that experience (that data) to generate a theory that paving roads makes insurgencies worse. This is inductive reasoning where we take one observation and generalize the observation to the world. We take our theory that paving roads worsens an insurgency, and we make a prediction about what the data will show in other insurgencies. If what I observed in Afghanistan is true of all insurgencies, I can make a prediction that I will see a similar relationship in other insurgencies like Egypt or Syria or Libya (deductive reasoning). The insurgency violence will be worse in those areas with paved roads. Ideally, I could do an experiment by randomly choosing some insurgency areas to be paved, and others not to be, and observe the effects at a later time.

This is a continuous cycle. Scientists go from data to theory to data to theory... And as the cycle continues, theories are refined, and they become better at explaining the world around us. We may find that our paved-roads theory needs to be modified. We might refine our theory by saying paved roads make the insurgency violence worse, but only in areas where it is not possible to travel across open ground easily. The revised theory, if it is a good revision, will be able to explain more data. The more observations we're able to explain with our theory the more useful that theory is.

This mechanism allows human knowledge to advance. Theories usually outlive their originators, and so it's through the revision of better theories that a discipline evolves and becomes better at what it does. The next generation of scholars and analysts can stand on our shoulders and advance the discipline. Otherwise, if we don't advance theory, if we don't stand on the shoulders of those that have come before us, the nature of wisdom stays one-generational. The future generation of analysts can only hope to be as good as the current generation of analysts.

An example of a data-strong / theory-strong approach in intelligence is Richard Cincotta's application of youth bulge theory. Youth bulge theory notes that the 16-to-30-year age range is associated with risk-taking, especially among males. Youth bulges occurring in this age group in developing countries are associated with higher unemployment and, as a result, a heightened risk of violence and political instability. So, youth bulge theory argues that an excess of young adult males predictably leads to social unrest, war and terrorism.

Cincotta wanted to add to this theory. He thought that youthful demographics would be more likely to have authoritarian regimes. After all, if you are a shop keeper in a country with large numbers of young, unemployed, risk-taking males, you may not really mind that the police have some extraordinary powers. But as a demography matures, these concerns fade and issues of freedom and civil rights become more important. What's interesting then is Cincotta made projections of future

populations, and identified which countries are most likely to go through this transitional phase from having an authoritarian style government, to having a form of government more like a liberal democracy [14]. He made this prediction based only on the demographic profiles. The first region that promised a shift to liberal democracy was a cluster along Africa's Mediterranean coast: Morocco, Algeria, Tunisia, Libya, and Egypt, none of which had ever experienced democracy in the recent past. We saw some of this transition potentially start during the Arab Spring. The other area is in South America: Ecuador, Columbia, and Venezuela, each of which attained liberal democracy demographically 'early' but was unable to sustain it. For the policymaker, these areas represent opportunities where US efforts stand a good chance of bringing more democratic forms of government. It also suggests that countries like Afghanistan are just too young demographically to see a democratic form of government anytime soon.

As theory becomes better at explaining the world around us, we may find we don't need to collect as much data as we used to. Theory can direct us to a few key pieces of information that will tell us most of what we want to know. If we can reach this stage, we transition to the high theory / low data quadrant, analogous to Snowden and Boone's simple context.

- 4) **High Reliance on Theory, Low Reliance on Data:** Mark Lowenthal [15] argues that the intelligence community has focused too much on text-based data manipulation and not enough on critical thinking, experience, and wisdom. Lowenthal doesn't claim that data are unimportant. However, as the title of his article ("A few knowledgeable men") suggests, you don't need huge numbers of people to look at all the data. Rather, you just need a smaller number of men and women who are knowledgeable about the background issues for a particular intelligence question. What you need is a few people who understand the underlying motivations, culture and history. If you just have a few knowledgeable men and women who understand theory, then you don't need to look at every piece of data. You understand the direction the wave of data is traveling even though you aren't examining every piece of data. Data is deemphasized in this approach.

I'd like to highlight a risk to this approach, however. We are good at creating a quick story to retroactively explain past results. An example I use in class is the earlier-mentioned theory about paving roads in Afghanistan. Sometimes in class I describe how we are actually helping the insurgency by paving roads. This makes a lot of sense to the students. Allied forces use a lot of air support and air transportation. The insurgents have to use ground transport. So, the roads are actually helping them more than they are helping us. Insurgents can now move more quickly and more efficiently move supplies and forces. In addition, allied forces are now using the roads a lot more too, and not relying on air transport, so it presents a concentrated target that the insurgency can take advantage of.

Sometimes I present the exact opposite finding in class. Paving roads in Afghanistan is hurting the insurgents. People are also able to easily explain these results as well. Well of course it hurts the insurgency! By building roads, you are building up the transportation infrastructure of the country. You're facilitating trade of goods. You're facilitating the growth of business which is going to help grow the economy of Afghanistan and give young Afghans options that they would not have had otherwise. Now there are things you can do with your life other than join the insurgency. The point here is that an idea that is not supported with data is just an idea. And many people can have conflicting ideas. The data should help you decide which opinion is correct.

If you have a well-established theory, and it is well supported, it can be used to make future predictions, or bring understanding. This is especially useful when data is not available or time is short. However, Snowden and Boone have a strong warning to organizations operating in the simple domain. The old rules work until they don't. They recommend challenging orthodoxy and avoiding complacency.

So, which of these views of knowledge is the correct one for intelligence analysis and for academic research? All of them have a place. Looking at the “snowflake” quadrant for a moment, we have to acknowledge that every event does have some aspects that are truly unique. For some events, truly nothing like them has ever been seen before or will ever be seen again. Amy Zegart gives the example of a nuclear Iran [16]. What would a nuclear Iran look like? What precedence should we look to in history for estimates on how a nuclear Iran might act? Only nine countries have nuclear weapons. Five got the bomb a very long time ago when the world climate was very different. Two of the more recent nations to acquire the bomb also seem like poor models: there’s North Korea, which hardly seems a generalizable model for anything, and South Africa, the only country that developed and then voluntarily dismantled its nuclear arsenal. Intelligence analysts don’t have a rich historical store of comparable cases to help assess future outcomes for Iran.

Data mining, even without theory, can play a vital role in intelligence in the area of indications and warnings, for instance. If we were to observe large amounts of activity over a very wide geographic area, and then create a picture of what normal activity looks like, then we can identify and highlight non-normal activity when it occurs. That could be a significant tool for warning.

A few knowledgeable men or women who know about theory can be very helpful when data is sparse or when time is short. The whole purpose of theory is to be able to give us the ability to understand the relationships in a broad sense without having to look at reams and reams of data. If our theories have been well validated, then a few knowledgeable men and women are very useful in understanding the world around us.

There is a rich history of applying the scientific method to both data and theory. We have this wonderful potential of building theory over time such that future generations of intelligence analysts can do a better job than we are doing today. That’s very powerful, but it also relies on some assumptions. It assumes that some relationships are going to be generalizable from situation to situation. That is to say, it assumes that there is the possibility of developing theory. It also assumes we will have data that we are capable of observing. Sometimes historians debate the circumstances surrounding particular events for decades. Sometimes data is not readily observable.

- 5) **How systems change over time:** The power of the theory-data framework comes into play when systemic change to the nature of a problem over time is considered. Let’s assume the environment we are operating in has undergone a “Black Swan” shock of some sort. Examples of this kind of shock might be Iraq invading Kuwait, or 9/11. We find ourselves in the chaotic quadrant of low data / low theory approaches. This is not the time for high data approaches such as data mining, or inferential statistics. The commander in this instance must act quickly with little information to stabilize the situation, stem further bleeding, and attempt to bring order. For example, after the invasion of Kuwait, US fighters were sent to sit on the tarmacs of Saudi Arabia to dissuade Iraq from continuing their invasion into Saudi Arabia. After 9/11, all commercial air traffic was grounded.

Once the environment stabilizes somewhat, we may find we have progressed to the high data / low theory quadrant of complexity. Different actors are manoeuvring for power and changes can have large, non-linear effects on the overall system. In this instance, the analyst is best served by high data approaches such as Social Network Analysis, which is useful in identifying who the major players are in the current system. Data mining helps us understand the new system we find ourselves in. Theory is hard to apply in the frequently changing dynamics of the current system. Arguably, many intelligence questions remain in this realm.

Perhaps, over time, the environment proceeds to a state of more stable institutions and power bases, rendering the environment a bit more predictable and allowing theory to make predictions over a

wider variety of observations. We find ourselves operating in the complicated realm, where high data / high theory approaches will be helpful. Linear regressions can be used to make consistent prediction of how much wheat Country A will produce, based on the institution's stated goals and a few environmental factors. Number of tanks produced can be estimated, based on imagery of production facilities and orders of raw materials. Experts are helpful in this realm, and high data / high theory techniques that help to aggregate expert opinion, such as Bayesian Belief Networks or Prediction Markets, may be helpful here.

Finally, certain problems may allow us to proceed to the last high theory / low data approach. In this quadrant, the problem is well understood, and we can rely on what has happened in the past to make good predictions about the future. Key pieces of data can be accessed, rather than attempting to look at all data. For instance, Country A has conducted many weapon tests in the past. We know activity at a few key buildings precipitates the next test, and we can reference those few buildings to predict future tests. This is the realm of "a few knowledgeable men" who have the background to know what to look for. They are able to make accurate predictions by looking at perhaps just a few key diagnostic pieces of information. ACH might work better here, since we can pre-populate a matrix with the key evidence we know to look for. Stable and predictive Indications and Warnings matrices can be developed.

In Snowden and Boone's Cynefin Framework, they warn about the simple/chaotic boundary. When operating in the simple domain it is easy to let complacency set in and the environment may change without our notice. In our example of predicting weapon testing, imagine new R&D buildings have been constructed that we are not aware of, and weapons testing has transitioned to those buildings. The old theories that worked in the simple domain no longer apply. We are launched into the chaotic realm to start over.

I'm not certain which SATs will work best in a particular quadrant. I do have some intuitions. If the domain of your intelligence question has become chaotic, I suspect it's not the right time for a linear regression. Save classic statistical analysis for domains in the complicated context. If you are in the complex domain, strategies that locate temporary patterns to exploit will be more helpful than attempting to validate theories. If your intelligence problem is in the simple domain, this might be a warning sign that complacency has crept in, and you should take measures to look at new data. My hope is that presentation of this framework provides vocabulary to discuss appropriate analytic focus for a given intelligence question.

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